Biodiversity — Statistical Modeling Comparison (The Beaches vs High Park-Swansea)

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1. Load inputs

```
# iNaturalist: load all exports we can find and combine
inat_files <- find_all("observations(-[0-9]+)?\\.csv$")
inat_files <- inat_files[!grepl("fields\\.csv$", inat_files, ignore.case = TRUE)]
if (!length(inat_files)) stop("No iNaturalist CSVs found. Expected files like observations-670364.csv")

tibble::tibble(iNat_file = basename(inat_files), path = inat_files) |>
  kbl3(caption = "Detected iNaturalist exports")
```

Table 1: Detected iNaturalist exports

iNat_file	path
observations-624727.csv	/home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/observations-624727.csv
observations-631589.csv	/home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/observations-631589.csv
observations-670364.csv	/home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/observations-670364.csv

```
paths <- list(
  neigh   = find_vector("(Neighbourhoods.*4326)", required = TRUE),
```

```r
  green   = find_vector("Green\\s*Spaces.*4326", required = TRUE),
  hp_trees = find_vector("High_Park_Tree", required = FALSE)
)

paths
```

```
## $neigh
## [1] "/home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/Neighbourhoods - 4326.shp"
##
## $green
## [1] "/home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/Green Spaces - 4326.shp"
##
## $hp_trees
## [1] "/home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/High_Park_Tree.shp"
```

```r
cat("High Park tree layer:", paths$hp_trees, "\n")
```

```
## High Park tree layer: /home/rstudio/STA490-Beach-BIA/EDA/Biodiversity/Dataset/High_Park_Tree.shp
```

```r
guess_col <- function(df, candidates){
  cand <- intersect(candidates, names(df))
  if (length(cand)) cand[1] else NA_character_
}

read_one_inat <- function(f) {
  df <- readr::read_csv(f, show_col_types = FALSE) |> janitor::clean_names()

  lat_col <- guess_col(df, c("latitude","lat","y","wei_du"))
  lon_col <- guess_col(df, c("longitude","lon","long","x","jing_du"))
  date_col <- guess_col(df, c("observed_on","observed_on_string","time_observed_at","date","observed_on
  sci_col  <- guess_col(df, c("scientific_name","taxon_name","name","xue_ming","scientific"))
  com_col  <- guess_col(df, c("common_name","com_name","common"))
  iconic_col <- guess_col(df, c("iconic_taxon_name","taxon_iconic_name","iconic_taxon","biao_zhi_lei_qu
  id_col <- guess_col(df, c("id","observation_id","obs_id"))

  if (any(is.na(c(lat_col, lon_col)))) stop(glue("Missing lat/lon columns in: {basename(f)}"))

  out <- df |>
    mutate(
      latitude  = suppressWarnings(as.numeric(.data[[lat_col]])),
      longitude = suppressWarnings(as.numeric(.data[[lon_col]])),
      date = if (!is.na(date_col)) as.Date(.data[[date_col]]) else NA_Date_,
      scientific_name = if (!is.na(sci_col)) as.character(.data[[sci_col]]) else NA_character_,
      common_name = if (!is.na(com_col)) as.character(.data[[com_col]]) else NA_character_,
      iconic_taxon_name = if (!is.na(iconic_col)) as.character(.data[[iconic_col]]) else NA_character_,
      observation_id = if (!is.na(id_col)) as.character(.data[[id_col]]) else NA_character_,
      source_file = basename(f)
    ) |>
    filter(!is.na(latitude), !is.na(longitude),
           between(latitude,-90,90), between(longitude,-180,180))

  out <- out |>
    mutate(
      category = case_when(
        str_detect(iconic_taxon_name, regex("Plantae",  TRUE)) ~ "Plants",
```

```
        str_detect(iconic_taxon_name, regex("Aves",     TRUE)) ~ "Birds",
        str_detect(iconic_taxon_name, regex("Insecta",  TRUE)) ~ "Insects",
        str_detect(iconic_taxon_name, regex("Mammalia", TRUE)) ~ "Mammals",
        str_detect(iconic_taxon_name, regex("Reptilia", TRUE)) ~ "Reptiles",
        str_detect(iconic_taxon_name, regex("Arachnida",TRUE)) ~ "Spiders",
        TRUE ~ "Other"
      )
    )

  out
}

inat <- purrr::map_dfr(inat_files, read_one_inat)

if (any(!is.na(inat$observation_id)) && any(nchar(inat$observation_id) > 0)) {
  inat <- inat |> distinct(observation_id, .keep_all = TRUE)
}

inat_sf <- st_as_sf(inat, coords = c("longitude","latitude"), crs = 4326, remove = FALSE)
inat_sf <- inat_sf |> filter(!is.na(date), date >= START_DATE)

inat |> count(source_file, sort = TRUE) |> kbl3(caption = "Observations per iNat export (after cleaning
```

Table 2: Observations per iNat export (after cleaning)

source_file	n
observations-631589.csv	74864
observations-670364.csv	37049
observations-624727.csv	3632

```
neigh <- safe_read_sf(paths$neigh)
green <- safe_read_sf(paths$green)

hp_trees <- if (!is.na(paths$hp_trees)) {
  tryCatch(safe_read_sf(paths$hp_trees), error = function(e) NULL)
} else NULL

# Find neighbourhood polygons
pat_beaches  <- regex("(^|\\b)the\\s*beaches(\\b|$)", ignore_case = TRUE)
pat_highpark <- regex("high\\s*park", ignore_case = TRUE) # should catch High Park-Swansea

beaches_poly <- find_polygon_by_name(neigh, pat_beaches)
hp_poly      <- find_polygon_by_name(neigh, pat_highpark)

if (is.null(beaches_poly) || nrow(beaches_poly) == 0) stop("Could not locate neighbourhood polygon for
if (is.null(hp_poly) || nrow(hp_poly) == 0) stop("Could not locate neighbourhood polygon for 'High Park

beaches_poly <- make_single_poly(beaches_poly)
hp_poly      <- make_single_poly(hp_poly)

# Areas (km^2)
area_tbl <- tibble::tibble(
```

```
  neighbourhood = c("The Beaches", "High Park-Swansea"),
  area_km2 = c(
    as.numeric(st_area(st_transform(beaches_poly, AREA_CRS))) / 1e6,
    as.numeric(st_area(st_transform(hp_poly, AREA_CRS))) / 1e6
  )
)
area_tbl |> kbl3(caption = "Neighbourhood areas (km²)")
```

Table 3: Neighbourhood areas (km²)

neighbourhood	area_km2
The Beaches	3.596
High Park-Swansea	7.240

2. Build modeling dataset

2.1 Clip observations to the two neighbourhoods

```
inat_beaches <- suppressWarnings(st_filter(inat_sf, sanitize_poly(beaches_poly))) |>
  mutate(neighbourhood = "The Beaches")

inat_hp <- suppressWarnings(st_filter(inat_sf, sanitize_poly(hp_poly))) |>
  mutate(neighbourhood = "High Park-Swansea")

inat_2n <- bind_rows(inat_beaches, inat_hp)

inat_2n |> st_drop_geometry() |>
  summarise(
    n_obs = n(),
    n_species = n_distinct(na.omit(scientific_name)),
    min_date = min(date),
    max_date = max(date)
  ) |> kbl3(caption = "Combined dataset (2 neighbourhoods) - quick check")
```

Table 4: Combined dataset (2 neighbourhoods) — quick check

n_obs	n_species	min_date	max_date
37496	2952	2020-01-01	2026-01-01

2.2 Define "green" indicator

- **The Beaches:** point is "green" if it falls inside the Toronto **Green Spaces** polygons (clipped to The Beaches).
- **High Park-Swansea:** point is "green" if it is within 30 meters of any point in **High_Park_Tree**.

```
# Beaches green polygons
beaches_green <- tryCatch({
  suppressWarnings(st_intersection(sanitize_poly(green), sanitize_poly(beaches_poly))) |> sanitize_poly
}, error = function(e) NULL)

beaches_green_m <- if (!is.null(beaches_green) && nrow(beaches_green) > 0) {
```

```r
  st_union(st_make_valid(st_transform(beaches_green, AREA_CRS)))
} else NULL

# High Park tree points in UTM
hp_trees_m <- if (!is.null(hp_trees) && nrow(hp_trees) > 0) st_transform(hp_trees, AREA_CRS) else NULL

# classify observations
inat_m <- st_transform(inat_2n, AREA_CRS)

is_green <- rep(FALSE, nrow(inat_m))

# Beaches: intersects green polygons
idx_b <- which(inat_m$neighbourhood == "The Beaches")
if (!is.null(beaches_green_m) && length(idx_b) > 0) {
  is_green[idx_b] <- lengths(st_intersects(inat_m[idx_b, ], beaches_green_m)) > 0
}

# High Park: within distance of High_Park_Tree points
idx_h <- which(inat_m$neighbourhood == "High Park-Swansea")
if (length(idx_h) > 0) {
  if (is.null(hp_trees_m) || nrow(hp_trees_m) == 0) {
    warning("High_Park_Tree layer not found. High Park 'green' indicator will be FALSE for all points.")
    is_green[idx_h] <- FALSE
  } else {
    is_green[idx_h] <- lengths(st_is_within_distance(inat_m[idx_h, ], hp_trees_m, dist = HP_TREE_GREEN_
  }
}

inat_2n <- inat_2n |>
  mutate(
    in_green = is_green,
    zone = ifelse(in_green, "Green", "Non-green"),
    year = lubridate::year(date),
    month = lubridate::month(date, label = TRUE, abbr = TRUE)
  )

inat_2n |> st_drop_geometry() |>
  count(neighbourhood, zone) |>
  arrange(neighbourhood, desc(n)) |>
  kbl3(caption = "Counts of observations classified as Green vs Non-green")
```

Table 5: Counts of observations classified as Green vs Non-green

neighbourhood	zone	n
High Park-Swansea	Green	31892
High Park-Swansea	Non-green	1272
The Beaches	Green	2666
The Beaches	Non-green	1666

3. Modeling 1 — Monthly sightings intensity (rate per km²)

We aggregate iNaturalist observations by **month** and fit a **Negative Binomial** model with an **offset** for neighbourhood area.

```
monthly <- inat_2n |>
  st_drop_geometry() |>
  group_by(neighbourhood, year, month) |>
  summarise(
    sightings = n(),
    unique_species = n_distinct(na.omit(scientific_name)),
    .groups = "drop"
  ) |>
  left_join(area_tbl, by = "neighbourhood")

monthly |> arrange(neighbourhood, year, month) |> head(20) |>
  kbl3(caption = "Monthly aggregation (first 20 rows)")
```

Table 6: Monthly aggregation (first 20 rows)

neighbourhood	year	month	sightings	unique_species	area_km2
High Park-Swansea	2020	Jan	17	9	7.24
High Park-Swansea	2020	Feb	42	31	7.24
High Park-Swansea	2020	Mar	118	58	7.24
High Park-Swansea	2020	Apr	234	123	7.24
High Park-Swansea	2020	May	408	237	7.24
High Park-Swansea	2020	Jun	1008	544	7.24
High Park-Swansea	2020	Jul	1106	532	7.24
High Park-Swansea	2020	Aug	680	342	7.24
High Park-Swansea	2020	Sep	519	284	7.24
High Park-Swansea	2020	Oct	231	120	7.24
High Park-Swansea	2020	Nov	148	89	7.24
High Park-Swansea	2020	Dec	32	22	7.24
High Park-Swansea	2021	Jan	54	33	7.24
High Park-Swansea	2021	Feb	129	83	7.24
High Park-Swansea	2021	Mar	288	69	7.24
High Park-Swansea	2021	Apr	764	182	7.24
High Park-Swansea	2021	May	946	270	7.24
High Park-Swansea	2021	Jun	575	279	7.24
High Park-Swansea	2021	Jul	824	349	7.24
High Park-Swansea	2021	Aug	814	358	7.24

```
# Model: sightings ~ neighbourhood + month + year + offset(log(area_km2))
fit_rate <- MASS::glm.nb(
  sightings ~ neighbourhood + month + year + offset(log(area_km2)),
  data = monthly
)

summary(fit_rate)
```

```
##
## Call:
## MASS::glm.nb(formula = sightings ~ neighbourhood + month + year +
##     offset(log(area_km2)), data = monthly, init.theta = 4.474918513,
```

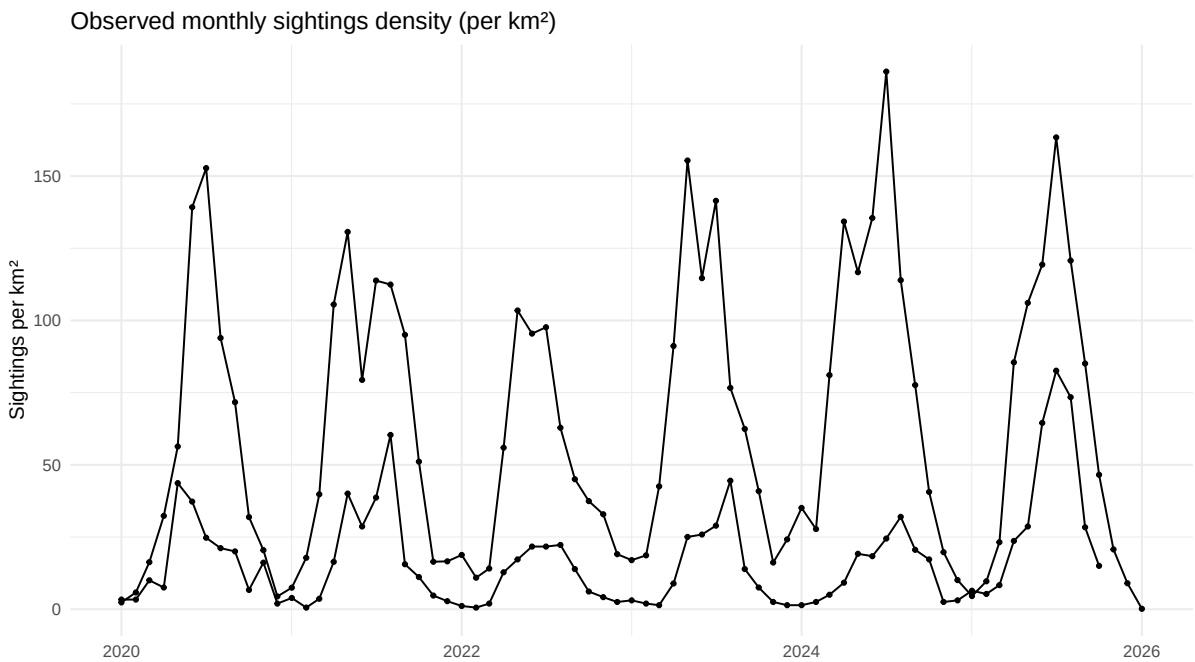
```
##     link = log)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -113.25052   48.59288  -2.331  0.01977 *
## neighbourhoodThe Beaches   -1.44014    0.08281 -17.391  < 2e-16 ***
## month.L                     0.39136    0.14735   2.656  0.00791 **
## month.Q                    -3.12513    0.14574 -21.443  < 2e-16 ***
## month.C                    -0.22194    0.14412  -1.540  0.12356
## month^4                     0.63905    0.14456   4.421 9.84e-06 ***
## month^5                    -0.25170    0.14492  -1.737  0.08243 .
## month^6                     0.19627    0.14440   1.359  0.17409
## month^7                    -0.14192    0.14292  -0.993  0.32071
## month^8                    -0.12150    0.14129  -0.860  0.38984
## month^9                     0.08935    0.13988   0.639  0.52300
## month^10                    0.10628    0.13874   0.766  0.44366
## month^11                   -0.03337    0.13809  -0.242  0.80904
## year                        0.05789    0.02403   2.409  0.01598 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(4.4749) family taken to be 1)
##
##     Null deviance: 851.88  on 142  degrees of freedom
## Residual deviance: 153.89  on 129  degrees of freedom
## AIC: 1576.4
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  4.475
##           Std. Err.:  0.578
##
##  2 x log-likelihood:  -1546.430
```

```r
coef_tbl <- as.data.frame(summary(fit_rate)$coefficients)
coef_tbl <- tibble::tibble(
  term = rownames(coef_tbl),
  estimate = coef_tbl[,1],
  se = coef_tbl[,2],
  z = coef_tbl[,3],
  p = coef_tbl[,4]
) |>
  mutate(
    IRR = exp(estimate),
    IRR_low = exp(estimate - 1.96*se),
    IRR_high = exp(estimate + 1.96*se)
  )

coef_tbl |> kbl3(caption = "Model 1 - Incidence Rate Ratios (IRR) for monthly sightings per km²")
```

Table 7: Model 1 — Incidence Rate Ratios (IRR) for monthly sightings per km²

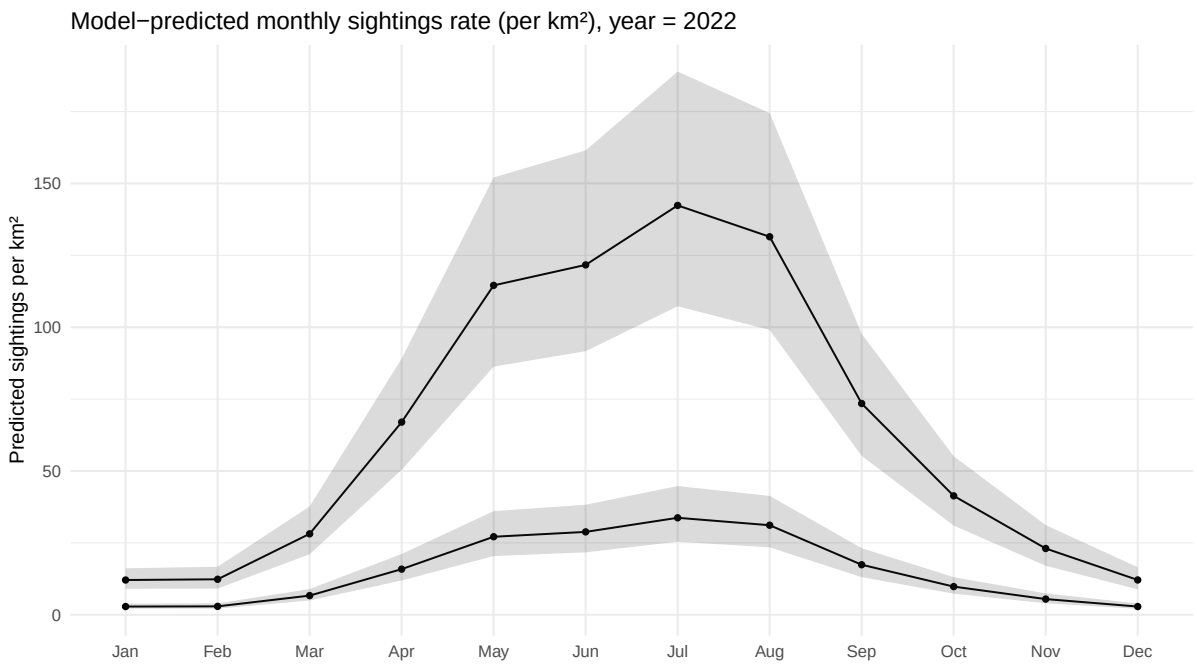
term	estimate	se	z	p	IRR	IRR_low	IRR_high
(Intercept)	-113.251	48.593	-2.331	0.020	0.000	0.000	0.000
neighbourhoodThe Beaches	-1.440	0.083	-17.391	0.000	0.237	0.201	0.279
month.L	0.391	0.147	2.656	0.008	1.479	1.108	1.974
month.Q	-3.125	0.146	-21.443	0.000	0.044	0.033	0.058
month.C	-0.222	0.144	-1.540	0.124	0.801	0.604	1.062
month^4	0.639	0.145	4.421	0.000	1.895	1.427	2.515
month^5	-0.252	0.145	-1.737	0.082	0.777	0.585	1.033
month^6	0.196	0.144	1.359	0.174	1.217	0.917	1.615
month^7	-0.142	0.143	-0.993	0.321	0.868	0.656	1.148
month^8	-0.121	0.141	-0.860	0.390	0.886	0.671	1.168
month^9	0.089	0.140	0.639	0.523	1.093	0.831	1.438
month^10	0.106	0.139	0.766	0.444	1.112	0.847	1.460
month^11	-0.033	0.138	-0.242	0.809	0.967	0.738	1.268
year	0.058	0.024	2.409	0.016	1.060	1.011	1.111

```r
monthly <- monthly |>
  mutate(
    date_month = as.Date(paste(year, as.integer(month), "01", sep = "-")),
    density_per_km2 = sightings / area_km2
  )

p_raw <- ggplot(monthly, aes(x = date_month, y = density_per_km2, group = neighbourhood)) +
  geom_line() +
  geom_point(size = 0.9) +
  labs(
    title = "Observed monthly sightings density (per km²)",
    x = NULL, y = "Sightings per km²"
  ) +
  theme_minimal()

p_raw
```




```
ggsave(file.path(FIG_DIR, "monthly_sightings_density_raw.png"), p_raw, width = 10, height = 5, dpi = 300
```

```
# Predicted monthly rate for a "typical" year (median of observed years)
typ_year <- median(monthly$year, na.rm = TRUE)

newdat <- expand.grid(
  neighbourhood = unique(monthly$neighbourhood),
  month = levels(monthly$month),
  year = typ_year,
  area_km2 = 1  # so predictions are "per km²"
)

pred <- predict(fit_rate, newdata = newdat, type = "link", se.fit = TRUE)

newdat$rate_per_km2 <- exp(pred$fit)
newdat$low <- exp(pred$fit - 1.96 * pred$se.fit)
newdat$high <- exp(pred$fit + 1.96 * pred$se.fit)

p_pred <- ggplot(newdat, aes(x = month, y = rate_per_km2, group = neighbourhood)) +
  geom_line() +
  geom_point(size = 1.2) +
  geom_ribbon(aes(ymin = low, ymax = high), alpha = 0.18, linetype = 0) +
  labs(
    title = glue("Model-predicted monthly sightings rate (per km²), year = {typ_year}"),
    x = NULL, y = "Predicted sightings per km²"
  ) +
  theme_minimal()

p_pred
```




```
ggsave(file.path(FIG_DIR, "model_pred_monthly_rate.png"), p_pred, width = 10, height = 5, dpi = 300)
```

4. Modeling 2 — Monthly species richness (adjusted for observation effort)

Raw species counts tend to increase with the number of observations. So we model **unique species per month** using an offset of `log(sightings)` (a simple proxy for sampling effort).

```
monthly2 <- monthly |>
  filter(sightings > 0) |>
  mutate(month = factor(month, levels = levels(month)))

fit_rich <- MASS::glm.nb(
  unique_species ~ neighbourhood + month + year + offset(log(sightings)),
  data = monthly2
)

summary(fit_rich)
```

```
##
## Call:
## MASS::glm.nb(formula = unique_species ~ neighbourhood + month +
##     year + offset(log(sightings)), data = monthly2, init.theta = 28.40276548,
##     link = log)
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)              60.397790  23.279968   2.594 0.009475 **
## neighbourhoodThe Beaches  0.598841   0.041054  14.587  < 2e-16 ***
## month.L                   0.189626   0.081022   2.340 0.019262 *
## month.Q                   0.161370   0.078343   2.060 0.039419 *
## month.C                  -0.258461   0.075978  -3.402 0.000669 ***
```

```
## month^4                  -0.003387   0.075712  -0.045 0.964319
## month^5                   0.054454   0.074618   0.730 0.465528
## month^6                  -0.091751   0.073165  -1.254 0.209826
## month^7                   0.172542   0.071039   2.429 0.015148 *
## month^8                   0.019584   0.068410   0.286 0.774663
## month^9                  -0.010309   0.065384  -0.158 0.874717
## month^10                 -0.024025   0.062130  -0.387 0.698984
## month^11                  0.054657   0.059896   0.913 0.361486
## year                     -0.030287   0.011510  -2.631 0.008506 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(28.4028) family taken to be 1)
##
##     Null deviance: 374.81  on 142  degrees of freedom
## Residual deviance: 132.07  on 129  degrees of freedom
## AIC: 1193.1
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  28.40
##           Std. Err.:  4.79
##
##  2 x log-likelihood:  -1163.123
```

```r
coef_rich <- as.data.frame(summary(fit_rich)$coefficients)
coef_rich <- tibble::tibble(
  term = rownames(coef_rich),
  estimate = coef_rich[,1],
  se = coef_rich[,2],
  z = coef_rich[,3],
  p = coef_rich[,4]
) |>
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96*se),
    RR_high = exp(estimate + 1.96*se)
  )

coef_rich |> kbl3(caption = "Model 2 - Rate Ratios for monthly unique species (offset = log(sightings))
```

Table 8: Model 2 — Rate Ratios for monthly unique species (offset = log(sightings))

term	estimate	se	z	p	RR	RR_low	RR_high
(Intercept)	60.398	23.280	2.594	0.009	1.699914e+26	2594885.424	1.113617e+46
neighbourhoodThe Beaches	0.599	0.041	14.587	0.000	1.820000e+00	1.679	1.973000e+00
month.L	0.190	0.081	2.340	0.019	1.209000e+00	1.031	1.417000e+00
month.Q	0.161	0.078	2.060	0.039	1.175000e+00	1.008	1.370000e+00
month.C	-0.258	0.076	-3.402	0.001	7.720000e-01	0.665	8.960000e-01

term	estimate	se	z	p	RR	RR_low	RR_high
month^4	-0.003	0.076	-0.045	0.964	9.970000e-01	0.859	1.156000e+00
month^5	0.054	0.075	0.730	0.466	1.056000e+00	0.912	1.222000e+00
month^6	-0.092	0.073	-1.254	0.210	9.120000e-01	0.790	1.053000e+00
month^7	0.173	0.071	2.429	0.015	1.188000e+00	1.034	1.366000e+00
month^8	0.020	0.068	0.286	0.775	1.020000e+00	0.892	1.166000e+00
month^9	-0.010	0.065	-0.158	0.875	9.900000e-01	0.871	1.125000e+00
month^10	-0.024	0.062	-0.387	0.699	9.760000e-01	0.864	1.103000e+00
month^11	0.055	0.060	0.913	0.361	1.056000e+00	0.939	1.188000e+00
year	-0.030	0.012	-2.631	0.009	9.700000e-01	0.949	9.920000e-01

5. Modeling 3 — Are observations more likely to occur in green areas?

This is an **observation-level** logistic regression:

- Response: `in_green` (Green vs Non-green)
- Predictors: neighbourhood, month, category, year

```r
obs_df <- inat_2n |>
  st_drop_geometry() |>
  filter(!is.na(month), !is.na(year)) |>
  mutate(
    month = factor(month, levels = levels(inat_2n$month)),
    category = factor(category),
    neighbourhood = factor(neighbourhood)
  )

fit_green <- glm(
  in_green ~ neighbourhood + month + category + year,
  data = obs_df,
  family = binomial()
)

summary(fit_green)
```

```
##
## Call:
## glm(formula = in_green ~ neighbourhood + month + category + year,
##     family = binomial(), data = obs_df)
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)               9.831971  24.774669   0.397  0.69147
## neighbourhoodThe Beaches -2.918523   0.044971 -64.898  < 2e-16 ***
## month.L                   0.349413   0.124612   2.804  0.00505 **
## month.Q                  -0.625482   0.119061  -5.253 1.49e-07 ***
## month.C                  -0.143771   0.108205  -1.329  0.18395
```

```
## month^4                  -0.174008   0.109802  -1.585  0.11303
## month^5                   0.017075   0.109872   0.155  0.87650
## month^6                  -0.080308   0.106029  -0.757  0.44880
## month^7                   0.322361   0.098166   3.284  0.00102 **
## month^8                  -0.007398   0.087793  -0.084  0.93284
## month^9                   0.181922   0.075999   2.394  0.01668 *
## month^10                  0.098820   0.064545   1.531  0.12576
## month^11                  0.050063   0.056419   0.887  0.37489
## categoryInsects           0.519740   0.060431   8.601  < 2e-16 ***
## categoryMammals           0.548415   0.104945   5.226 1.73e-07 ***
## categoryOther             0.642225   0.118204   5.433 5.53e-08 ***
## categoryPlants            0.726072   0.063757  11.388  < 2e-16 ***
## categoryReptiles          0.255908   0.196478   1.302  0.19275
## categorySpiders           0.073068   0.178445   0.409  0.68219
## year                     -0.003472   0.012248  -0.283  0.77680
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20603  on 37495  degrees of freedom
## Residual deviance: 16258  on 37476  degrees of freedom
## AIC: 16298
##
## Number of Fisher Scoring iterations: 6
```

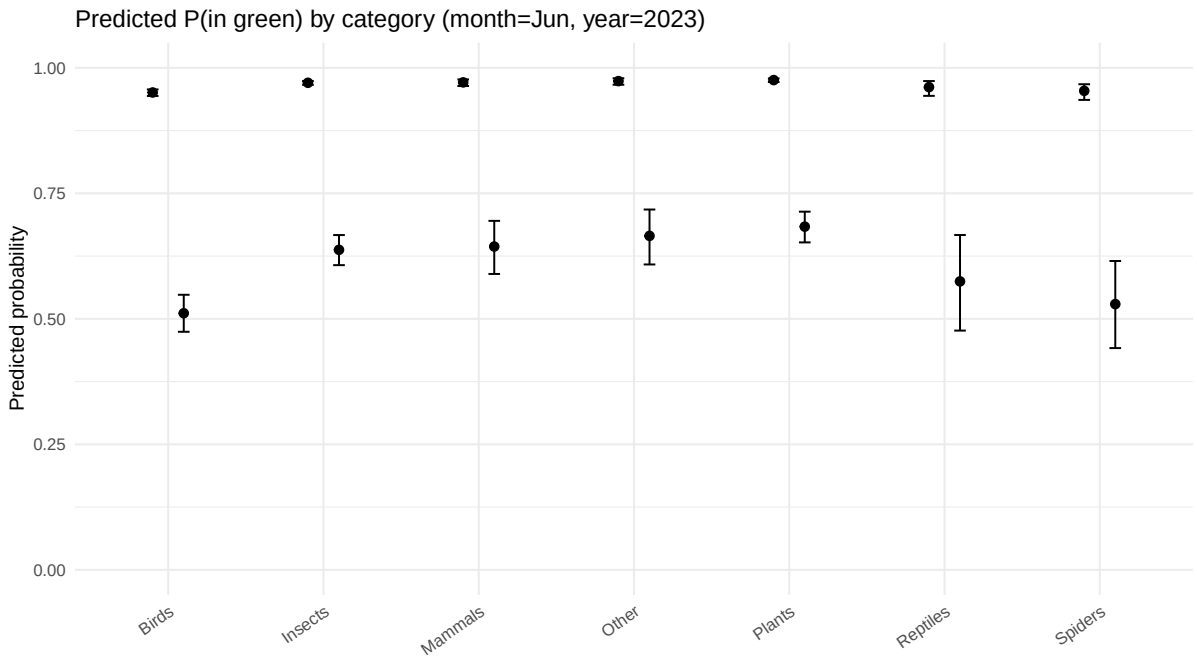
```r
# Odds ratios
coef_g <- as.data.frame(summary(fit_green)$coefficients)
coef_g <- tibble::tibble(
  term = rownames(coef_g),
  estimate = coef_g[,1],
  se = coef_g[,2],
  z = coef_g[,3],
  p = coef_g[,4]
) |>
  mutate(
    OR = exp(estimate),
    OR_low = exp(estimate - 1.96*se),
    OR_high = exp(estimate + 1.96*se)
  )

coef_g |> kbl3(caption = "Model 3 - Odds Ratios for being in Green areas (logistic regression)")
```

Table 9: Model 3 — Odds Ratios for being in Green areas (logistic regression)

term	estimate	se	z	p	OR	OR_low	OR_high
(Intercept)	9.832	24.775	0.397	0.691	18619.613	0.000	2.283466e+25
neighbourhoodThe Beaches	-2.919	0.045	-64.898	0.000	0.054	0.049	5.900000e-02
month.L	0.349	0.125	2.804	0.005	1.418	1.111	1.811000e+00
month.Q	-0.625	0.119	-5.253	0.000	0.535	0.424	6.760000e-01
month.C	-0.144	0.108	-1.329	0.184	0.866	0.701	1.071000e+00
month^4	-0.174	0.110	-1.585	0.113	0.840	0.678	1.042000e+00

term	estimate	se	z	p	OR	OR_low	OR_high
month^5	0.017	0.110	0.155	0.876	1.017	0.820	1.262000e+00
month^6	-0.080	0.106	-0.757	0.449	0.923	0.750	1.136000e+00
month^7	0.322	0.098	3.284	0.001	1.380	1.139	1.673000e+00
month^8	-0.007	0.088	-0.084	0.933	0.993	0.836	1.179000e+00
month^9	0.182	0.076	2.394	0.017	1.200	1.034	1.392000e+00
month^10	0.099	0.065	1.531	0.126	1.104	0.973	1.253000e+00
month^11	0.050	0.056	0.887	0.375	1.051	0.941	1.174000e+00
categoryInsects	0.520	0.060	8.601	0.000	1.682	1.494	1.893000e+00
categoryMammals	0.548	0.105	5.226	0.000	1.731	1.409	2.126000e+00
categoryOther	0.642	0.118	5.433	0.000	1.901	1.508	2.396000e+00
categoryPlants	0.726	0.064	11.388	0.000	2.067	1.824	2.342000e+00
categoryReptiles	0.256	0.196	1.302	0.193	1.292	0.879	1.898000e+00
categorySpiders	0.073	0.178	0.409	0.682	1.076	0.758	1.526000e+00
year	-0.003	0.012	-0.283	0.777	0.997	0.973	1.021000e+00

```r
# Predicted probability of "in_green" by category, holding month/year at typical values
typ_year2 <- median(obs_df$year, na.rm = TRUE)
typ_month <- levels(obs_df$month)[6]  # ~June

nd <- expand.grid(
  neighbourhood = levels(obs_df$neighbourhood),
  category = levels(obs_df$category),
  month = typ_month,
  year = typ_year2
)

pr <- predict(fit_green, newdata = nd, type = "link", se.fit = TRUE)
nd$prob <- plogis(pr$fit)
nd$low  <- plogis(pr$fit - 1.96*pr$se.fit)
nd$high <- plogis(pr$fit + 1.96*pr$se.fit)

p_g <- ggplot(nd, aes(x = category, y = prob, group = neighbourhood)) +
  geom_point(position = position_dodge(width = 0.4), size = 2) +
  geom_errorbar(aes(ymin = low, ymax = high),
                position = position_dodge(width = 0.4), width = 0.15) +
  coord_cartesian(ylim = c(0,1)) +
  labs(
    title = glue("Predicted P(in green) by category (month={typ_month}, year={typ_year2})"),
    x = NULL, y = "Predicted probability"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))

p_g
```




```
ggsave(file.path(FIG_DIR, "pred_in_green_by_category.png"), p_g, width = 10, height = 5, dpi = 300)
```

6. Interpretation notes (for the report write-up)

- **Model 1 (sightings rate per km²):** The coefficient for *neighbourhood* compares overall observation intensity between The Beaches and High Park-Swansea, adjusting for seasonality (month) and long-run trend (year).
- **Model 2 (species richness adjusted for effort):** With `offset(log(sightings))`, the neighbourhood effect answers: "given similar observation volume, which neighbourhood tends to show higher monthly richness?"
- **Model 3 (green propensity):** Helps validate the "green-vs-non" framing: are observations systematically concentrated in green areas, and does that differ by neighbourhood/category?

Limitations to mention (quick bullets for Discussion):

- iNaturalist data reflect **observer effort and accessibility**, not just ecological truth.
- "Green" is defined differently across neighbourhoods (polygon green spaces vs proximity to High Park tree points).
- Spatial autocorrelation is not explicitly modeled here (kept simple for interpretability).